OBTAINING THE COMMITTOR FUNCTION IN CRYO-EM

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Abstract. Cryogenic electron microscopy (cryo-EM) provides a powerful window into the conformational heterogeneity of biological macromolecules, but extracting physically meaningful dynamical information from conformational distributions remains challenging. Recent methods, such as RECOVAR, compute an estimate of the conformational density in a low-dimensional principal component analysis (PCA) latent space, which, by Boltzmann statistics, encodes an effective free energy landscape. We propose using this landscape to compute the *committor function* between pairs of conformational states identified by RECOVAR. The committor, satisfying a backward Kolmogorov equation, quantifies the probability that a macromolecule undergoing Langevin dynamics transitions to one state before another. Because the RECOVAR latent space is typically 4–10 dimensional, traditional numerical PDE solvers are intractable. We argue that the tensor train (TT) approach of [1] provides a natural remedy, enabling committor computation with complexity linear in the number of dimensions.

1. Introduction

Proteins and macromolecular complexes are not static objects but rather they populate a continuum of conformations, and the transitions between these conformations underlie virtually all biological functions. Cryo-EM, by imaging many individual particles in a vitrified state, offers snapshots of this conformational distribution. A central goal of cryo-EM heterogeneity analysis is therefore not merely to catalog conformational states, but to understand the dynamics between them.

In [2], Gilles and Singer present RECOVAR, which is a statistically rigorous approach to this problem of analyzing the heterogeneous distributions of conformations. It estimates a latent representation of the heterogeneous conformational distribution using a regularized covariance estimator and then reconstructs a conformational density in that latent space. This density gives access to stable states and the free energy landscape of the molecule via Boltzmann statistics. In this pipeline, this latent space gives us a reduced coordinate system in which the conformational landscape can be analyzed quantitatively, as we will soon see.

Once this latent density is obtained, a natural next object is the *committor function*. Put simply, the committor is the probability that a trajectory initialized at a given point reaches some state B before state A. For overdamped Langevin dynamics (which we assume the molecules in cryo-EM follow), the committor solved a backward Kolmogorov equation with Dirichlet boundary conditions on the two target sets. Naively, solving such a problem would have a computational and memory complexity that is exponential in dimension. In [1], however, Chen et al. demonstrate that this problem can be written in a (soft) variational form, which can then be solved efficiently by representing the density and the committor in *tensor train* (TT) form. From here, the committor can be obtained via a clever alternating least squares, resulting in both the computational and memory complexity scaling linearly with dimension.

Thus to apply such an algorithm in cryo-EM, one could imagine a general pipeline that would begin with using RECOVAR to infer a low-dimensional free-energy landscape in latent space. Then one would apply the tensor-train committor algorithm on that latent landscape to characterize the transition pathways between the conformational states. This would not only give us the stable states but also the transition-state structure of the molecule, as demonstrated in Section 4.2.

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2. The RECOVAR Latent space and effective free energy

2.1. The RECOVAR pipeline. RECOVAR starts from the cryo-EM image-formation model

$$y_i = P_i x_i + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \Lambda_i), \tag{1}$$

where y_i is the observed 2D image, x_i is the underlying 3D conformation, and P_i combines projection, pose, and CTF effects. In this model, they take the poses ϕ_i to already be estimated (using, e.g., a consensus reconstruction) so that, using a regularized least-squares procedure, they estimate the covariance matrix $\hat{\Sigma}$ from

$$\hat{\Sigma} := \arg\min_{\Sigma} \sum_{i=1}^{n} \|(y_i - P_i\hat{\mu})(y_i - P_i\hat{\mu})^* - (P_i \Sigma P_i^* + \Lambda_i)\|_F^2 + \|\Sigma\|_R^2, \tag{2}$$

where $\|\cdot\|_F^2$ is the Frobenius norm, $\|A\|_R^2 = \sum_{i,j} A_{i,j}^2 R_{i,j}$ for R the regularization weights, and $\hat{\mu}$ is the mean estimated from solving the linear least-squares problem

$$\hat{\mu} := \arg\min_{\mu} \sum_{i=1}^{n} \|y_i - P_i\mu\|_{\Lambda_i^{-1}}^2 + \|\mu\|_w^2 \quad \text{where } \|v\|_w^2 = \sum_i |v_i|^2 w_i \quad \& \quad \|v\|_{\Lambda_i^{-1}}^2 = v^* \Lambda_i^{-1} v. \tag{3}$$

The d-leading eigenvectors of $\hat{\Sigma} \in \mathbb{C}^{N^3 \times N^3}$ are then used to form $U \in \mathbb{C}^{N^3 \times d}$, defining the principal components. Hence we can embed each conformation x_i into latent coordinates as

$$z_i = U^*(x_i - \hat{\mu}) \in \mathbb{R}^d. \tag{4}$$

Further, because U is orthogonal, distance between conformations in volume space are preserved up to truncation error, i.e. $\|x_i - x_j\|^2 \approx \|z_i - z_j\|^2$.

2.2. Conformational density and free energy. These RECOVAR latent coordinates provide a low-dimensional space in which the conformational distribution of the dataset can be estimated directly. In practice, this is done by applying kernel density estimation (see Section 5 for further discussion on this) to the embedded samples $\{z_i\}_{i=1}^n$, yielding an empirical density $\hat{\rho}(z)$. This density is not itself a physical Boltzmann density in the strict sense, but it provides an effective description of how frequently different regions of latent space are occupied by the conformational ensemble represented in the dataset.

It is therefore natural to define an associated effective free energy landscape by

$$F(z) := -\beta^{-1} \log \hat{\rho}(z) + C, \tag{5}$$

where $\beta = 1/T$ and C is an arbitrary additive constant. Regions of high conformational density correspond to low values of F, while low-density regions correspond to energetic barriers separating distinct basins. In this sense, metastable conformational states appear as local minima of the effective free energy, and transitions between them occur through the intervening high-free-energy regions.

Once the conformational distribution has been reduced to a tractable latent density, one may pose the committor problem on this latent space, making this interpretation is particularly useful for our purposes. The resulting committor then quantifies the probability that a trajectory initiated at a given latent configuration first reaches one metastable basin rather than another, thereby giving a notion of how a molecule may progress through its conformational states.

3. Obtaining the committor function

3.1. What we need to solve. Recall that we are interested in the overdamped Langevin process

$$d\boldsymbol{X}_t = -\nabla V(\boldsymbol{X}_t)dt + \sqrt{2\beta^{-1}} d\boldsymbol{W}_t, \tag{6}$$

where $\boldsymbol{X}_t \in \Omega \subset \mathbb{R}^d$ is the state of the system, $V : \Omega \to \mathbb{R}$ is a smooth, potential energy function, $\beta = 1/T$ is the inverse of the temperature, and $\boldsymbol{W}_t$ is a d-dimensional Wiener process. For a potential function V that is confining on Ω, letting $Z_\beta = \int_\Omega \exp(-\beta V(\boldsymbol{x}))d\boldsymbol{x}$ be the partition function, one then sees that the equilibrium probability distribution of these dynamics is the Boltzmann-Gibbs distribution

$$p(\boldsymbol{x}) = \frac{1}{Z_\beta} \exp(-\beta V(\boldsymbol{x})). \tag{7}$$

As in the case of conformational heterogeneity in cryo-EM, we are interested in the transition between two simply connected domains $A, B \subset \Omega$ that have smooth boundaries and are disjoint. To investigate this, we define the committor function $q : \Omega \to [0, 1]$ as

$$q(\boldsymbol{x}) = \mathbb{P}(\tau_A < \tau_B : \boldsymbol{X}_0 = \boldsymbol{x}), \tag{8}$$

where τ_A and τ_B are the times it takes for our process to first hit A and B, respectively. Computing this function comes from solving the following backward Kolmogorov equation with Dirichlet boundary conditions

$$-\beta^{-1}\Delta q(\boldsymbol{x}) + \nabla V(\boldsymbol{x}) \cdot \nabla q(\boldsymbol{x}) = 0 \text{ in } \Omega\backslash(A \cup B),\ q(\boldsymbol{x})|_{\partial A} = 0,\ q(\boldsymbol{x})|_{\partial B} = 1. \tag{9}$$

A careful analysis (see, e.g., [3]) then shows that this can be rewritten as a variational problem, enforcing the boundary conditions as penalty terms

$$\arg\min_q \int_\Omega |\nabla q(\boldsymbol{x})|^2 p(\boldsymbol{x}) d\boldsymbol{x} + \rho \int_{\partial A} q(\boldsymbol{x})^2 p_{\partial A}(\boldsymbol{x}) d\boldsymbol{x} + \rho \int_{\partial B} (q(\boldsymbol{x}) - 1)^2 p_{\partial B}(\boldsymbol{x}) d\boldsymbol{x}, \tag{10}$$

where $p_{\partial A}$ and $p_{\partial B}$ are just p supported on ∂A and ∂B respectively. These boundary measures break the TT structure, however, because they are not absolutely continuous with respect to the Lesbesgue measure. This means that this formulation is not directly compatible with optimization even when one assumes q to have TT form. So, as in [1], we consider the *soft committor function*

$$\arg\min_q \int_\Omega |\nabla q(\boldsymbol{x})|^2 p(\boldsymbol{x}) d\boldsymbol{x} + \rho \int_\Omega q(\boldsymbol{x})^2 p_A(\boldsymbol{x}) d\boldsymbol{x} + \rho \int_\Omega (q(\boldsymbol{x}) - 1)^2 p_B(\boldsymbol{x}) d\boldsymbol{x}, \tag{11}$$

where p_A and p_B are probability densities that are absolutely continuous with respect to the Lebesgue measure on Ω. In what follows we will refer to soft committor functions as committor functions, as they admit similar probabilistic interpretations. For a more thorough analysis of this we refer the reader to Appendix A of [1].

3.2. **Reformulating this.** A key ingredient to solving this efficiently is the *tensor train* (TT) approximation defined as follows.

Definition 1. *Let $\mathcal{A} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$ be a d-way tensor with entries indexed by $(i_1, \ldots, i_d)$. We say that $\mathcal{A}$ is a TT with ranks $\boldsymbol{r} = (r_0, \ldots, r_d)$ where $r_0 = r_d = 1$ if one can write*

$$\mathcal{A}(i_1, \ldots, i_d) = \sum_{\alpha_0,\ldots,\alpha_d=1}^{r_0,\ldots,r_d} \mathcal{G}_1(\alpha_0, i_1, \alpha_1) \cdots \mathcal{G}_d(\alpha_{d-1}, i_d, \alpha_d) = \mathcal{G}_1(:, i_1, :) \cdots \mathcal{G}_d(:, i_d, :) \tag{12}$$

for all $(i_1, \ldots, i_d)$, where we note $\mathcal{G}_k(:, i_k, :) \in \mathbb{R}^{r_{k-1}\times r_k}$. The 3-way tensor $\mathcal{G}_k \in \mathbb{R}^{r_{k-1}\times n_k \times r_k}$ is called the k-th core of $\mathcal{A}$.

Following Section 3 of [1], we now obtain the committor function by solving eq. (11), assuming that q has a TT parametrization. We show that by approximating the equilibrium probability distribution p in this format, the optimization problem can be solved using a standard alternating least squares approach.

Writing our domain like $\Omega = \Omega_1 \times \cdots \Omega_d$, we let $\{\phi_j^{(k)}\}_{j=1}^\infty$ be an orthogonal basis for each $L^2(\Omega_k)$. To make this problem finite dimensional, we consider the subspace of $L^2(\Omega_k)$ spanned by the first $L^{(k)}$ basis functions. We then consider an expansion of q in this tensor product basis

$$q(\boldsymbol{x}) = \sum_{\boldsymbol{i}} \mathcal{Q}(\boldsymbol{i}) \phi_{i_1}^{(1)}(x_1) \cdots \phi_{i_d}^{(d)}(x_d), \tag{13}$$

where we defined $\boldsymbol{i} = (i_1, \ldots, i_d)$. Using this, we can rewrite the optimization problem as

(14)

$$\arg\min_{\mathcal{Q}} \sum_{k=1}^d \sum_{\boldsymbol{i},\boldsymbol{j}} H^k(\boldsymbol{i};\boldsymbol{j}) \mathcal{Q}(\boldsymbol{i}) \mathcal{Q}(\boldsymbol{j}) + \rho \sum_{\boldsymbol{i},\boldsymbol{j}} H^A(\boldsymbol{i};\boldsymbol{j}) \mathcal{Q}(\boldsymbol{i}) \mathcal{Q}(\boldsymbol{j}) + \rho \sum_{\boldsymbol{i},\boldsymbol{j}} H^B(\boldsymbol{i};\boldsymbol{j}) \mathcal{Q}(\boldsymbol{i}) \mathcal{Q}(\boldsymbol{j}) - 2\rho \sum_{\boldsymbol{i}} \mathcal{Q}(\boldsymbol{i}) h^B(\boldsymbol{i}) + \rho,$$

where the first and second terms equal the first and second terms of eq. (11) respectively, the last three terms equal the third term of eq. (11), and we defined

$$H^k(\boldsymbol{i};\boldsymbol{j}) = \int_\Omega \frac{\partial}{\partial x_k}\left[\phi^{(1)}_{i_1}(x_1)\cdots\phi^{(d)}_{i_d}(x_d)\right]\frac{\partial}{\partial x_k}\left[\phi^{(1)}_{j_1}(x_1)\cdots\phi^{(d)}_{j_d}(x_d)\right]p(\boldsymbol{x})d\boldsymbol{x} \tag{15}$$

$$H^{A,B}(\boldsymbol{i};\boldsymbol{j}) = \int_\Omega \phi^{(1)}_{i_1}(x_1)\cdots\phi^{(d)}_{i_d}(x_d)\phi^{(1)}_{j_1}(x_1)\cdots\phi^{(d)}_{j_d}(x_d)p_{A,B}(\boldsymbol{x})d\boldsymbol{x} \tag{16}$$

$$h^B(\boldsymbol{i}) = \int_\Omega \phi^{(1)}_{i_1}(x_1)\cdots\phi^{(d)}_{i_d}(x_d)p_B(\boldsymbol{x})d\boldsymbol{x}. \tag{17}$$

Computing these tensors explicitly is of course intractible, as it requires integration over $\Omega \subset \mathbb{R}^d$ and storing tensors of size exponential in d. We now show how we can use TT approximations to obtain these tensors and solve eq. (14) with computational and storage complexities that scale linearly in d.

3.3. **Constructing the tensors efficiently.** Let's begin by constructing H^k, assuming that the equilibrium density can be approximated in TT form, i.e.

$$p(\boldsymbol{x}) = \sum_{\substack{m_1,\dots,m_d\\ \alpha_0,\dots,\alpha_d}} \mathcal{P}_1(\alpha_0,m_1,\alpha_1)\cdots\mathcal{P}_d(\alpha_{d-1},m_d,\alpha_d)\psi^{(1)}_{m_1}(x_1)\cdots\psi^{(d)}_{m_d}(x_d), \tag{18}$$

where $\{\psi^{(k)}_j\}_{j=1}^{K^{(k)}}$ are univariate basis functions $\Omega_k \to \mathbb{R}$ and $\mathcal{P}_k \in \mathbb{R}^{r_{k-1}\times K^{(k)}\times r_k}$ is the coefficient tensor core for $k = 1,\dots,d$ (in general, if we say that a function has a TT form, we simply mean that it can be expressed in this way). Hence we can split up our d-dimensional integral into a sum and product of 1-dimensional integrals. More precisely, for $\ell = 1,\dots,d$, defining

$$A_\ell(\alpha_{\ell-1},\alpha_\ell) := \int_{\Omega_\ell} f_\ell(x_\ell)\mathcal{P}_\ell(\alpha_{\ell-1},m_\ell,\alpha_\ell)\psi^{(\ell)}_{m_\ell}(x_\ell)dx_\ell \quad \text{where } f_\ell(x_\ell) = \begin{cases}\phi^{(\ell)}_{i_\ell}(x_\ell)\,\phi^{(\ell)}_{j_\ell}(x_\ell), & \ell \neq k,\\ \frac{\partial\phi^{(\ell)}_{i_\ell}}{\partial x_\ell}\frac{\partial\phi^{(\ell)}_{j_\ell}}{\partial x_\ell}, & \ell = k,\end{cases} \tag{19}$$

we can write $H^k = A_1\cdots A_d$. Because the shape of these d matrices is independent of dimension, we see that this computation of H^k scales linearly in d. Many of these A_ℓ can clearly be reused for other k so it is not difficult to see that we can compute $\{H^k\}_{k=1}^d$ in time that scales linearly with the dimension. Letting $K := \max_\ell K^{(\ell)}$, $L := \max_\ell L^{(\ell)}$, and $r = \max_\ell r_\ell$, a careful analysis shows that in this form, $\{H^k\}_{k=1}^d$ can be computed with a computational cost of $\mathcal{O}(dr^2KL^2)$ and a memory complexity of $\mathcal{O}(dKL^2 + dr^2L^2)$ (including the storage of intermediary tensors).

The construction for H^A and H^B is essentially the same except now we are representing p_A and p_B in TT format. This representation is justified because in high dimensions the equilibrium density concentrates so that most of the probability mass lies on a thin shell and correlations between these shells are weak (for spheres this is captured by the Gaussian annulus theorem). As a result p_A and p_B retain low-rank TT approximations. For a more careful consideration of this, we refer the reader to Section 3.3 of [1]. From here we basically just apply the analysis done for H^k and letting s be the maximal rank of the cores used to represent p_A and K_A be the maximal number of basis modes, we obtain a time complexity of $\mathcal{O}(ds^2K_AL^2)$ to compute H^A. Defining similar things for B, we obtain a time complexity of $\mathcal{O}(dt^2K_BL^2)$ for computing H^B. The memory complexities are $\mathcal{O}(dK_AL^2+ds^2L^2)$ and $\mathcal{O}(dK_BL^2+dt^2L^2)$ respectively.

Lastly, using this TT approximation for p_B, we can obtain h^B in a similar way with computational and memory complexities of $\mathcal{O}(dt^2K_BL)$ and $\mathcal{O}(dK_BL + dt^2L)$ respectively. Now that all of these tensors are in TT form, we may turn to actually solving the minimization problem eq. (14).

3.4. **Optimization.** Before tackling this, we still must approximate the coefficient tensor in TT format as

$$\mathcal{Q}(\boldsymbol{i}) := \sum_{\alpha_0,\dots,\alpha_d} \mathcal{Q}_1(\alpha_0,i_1,\alpha_1)\cdots\mathcal{Q}_d(\alpha_{d-1},i_d,\alpha_d). \tag{20}$$

We can now solve the minimization from eq. (14) by an alternating least squares (ALS) approach. That is, in each ALS iteration, we loop over the dimensions $k = 1,\dots,d$. At step k, we treat all coefficient cores but $\mathcal{Q}_k$ as constant, giving us an unconstrained least squares problem for $\mathcal{Q}_k$. Naively, the computational complexity would be $\mathcal{O}(d^2)$ since we must sum d terms, each of which requires $\mathcal{O}(d)$ tensor contractions. Using the same trick as in the computation of H^k, the computational complexity can be brought down to $\mathcal{O}(d)$.

Hence, we obtain the key result of this algorithm: *using TT factorizations, the committor function can be computed by solving eq.* (14) *with a computational and memory complexity that scales linearly in the dimension.*

3.5. **Connecting this to RECOVAR.** The RECOVAR latent space is a very natural setting for this method. The embedding dimension is small enough that the committor problem is meaningful, but still large enough that direct PDE discretization is far too expensive. A tensor-train method therefore sits in exactly the right regime because it is flexible enough to represent nontrivial transition geometry, but structured enough to avoid the curse of dimensionality. In practice, an example pipeline could be as follows.

(1) Use RECOVAR to estimate the latent coordinates and density (maybe already in TT format, see Section 5).
(2) Define metastable sets A and B in that latent space.
(3) Construct a TT approximation of the density (maybe not needed) and the soft boundary measures.
(4) Solve the variational committor problem by ALS.
(5) Extract transition pathways and transition-state regions from the resulting committor.

4. Results

The code for these results can be found in the `quickstart.ipynb` notebook at `https://github.com/matias-ma/committor-tt`. We note that the implementation in this repository is a proof-of-concept and lacks full robustness. We also note that in the applications below, we found that many of the results are very sensitive to parameter selection. With a more careful parameter selection, some of the results below can likely be improved upon.

4.1. **Double-well.** As a sanity check for our implementation of the above algorithm, we consider a separable d-dimensional double-well model in which each coordinate experiences the same quartic potential

$$V(x) = (x^2 - 1)^2, \tag{21}$$

so that the full equilibrium density factorizes exactly as

$$p(\boldsymbol{x}) \propto \prod_{\ell=1}^{d} \exp\bigl(-\beta V(x_\ell)\bigr) = \exp\Bigl(-\beta \sum_{\ell=1}^{d} (x_\ell^2 - 1)^2\Bigr). \tag{22}$$

This is a particularly convenient test problem because the density has exact product structure, so the tensors appearing in the variational formulation admit low-rank TT approximations, yet the resulting committor still has a nontrivial transition layer between the two metastable basins.

We define the two metastable states to be sets that are small neighborhoods around the two wells,

$$\boldsymbol{x}_A = (-1, \dots, -1), \qquad \boldsymbol{x}_B = (+1, \dots, +1), \tag{23}$$

and replace the sharp indicators of A and B by smooth Gaussian product weights centered at these points (i.e. p_A and p_B are taken to be separable Gaussian densities that concentrate around $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$, respectively). In the code, the width parameter σ controls the size of these soft metastable neighborhoods. The parameter ρ plays the role of a penalty weight in the variational formulation, enforcing the boundary conditions associated with the committor more strongly as ρ increases.

We then solve the committor problem in two ways. First, we compute the full $d = 4$ TT approximation using a Legendre basis on the box $[-2, 2]^4$. Second, we solve the corresponding one-dimensional double-well problem on $[-2, 2]$ using a higher-resolution Legendre approximation, which serves as a reference solution. To compare the two, we evaluate the 4-dimensional committor along the diagonal slice $(x, x, x, x),\ x \in [-1, 1]$, and plot this against the one-dimensional reference committor.

This benchmark is useful for several reasons. First, it verifies that the TT solver reproduces the expected transition profile in a setting where the underlying density is completely separable. Second, because the 4D problem is already beyond the range of naive grid-based discretization, it demonstrates that the TT approach can solve a genuinely higher-dimensional committor problem at modest cost.

We note that these results are slightly different from those reported in [1] because we use different bases. In their example, they use a specialized basis tailored to the double-well experiment, while we use a simple Legendre basis. Nevertheless, the qualitative profile and the location of the transition region match their results rather well.

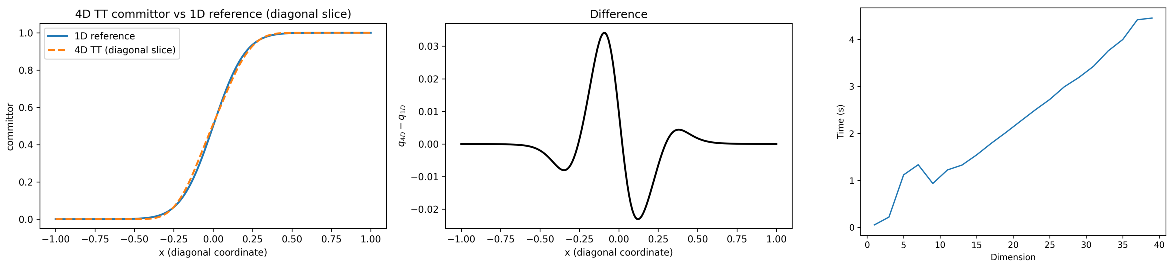

FIGURE 1. This example demonstrates that for a toy example this algorithm works as expected. We note that any differences between these results and the ones from Section 4.1 of [1] arise from the fact that they use a special basis and we just used a simple Legendre basis.

4.2. **Toy RECOVAR example.** Finding an interesting example for this was nontrivial without a tensor train approximation algorithm (i.e. you have to build the function from the ground up in tensor train form). Our primary goal was to come up with a conformational density and metastable states that look like they could come from a RECOVAR latent space embedding (see, e.g., Figure 3 of [2]). In this we had some success but there is still lots of room for improvement.

To construct a conformational density that minimally captures the essential features, we worked in a two-dimensional latent space $z = (z_1, z_2) \in [-1.5, 1.5]^2$ and define an equilibrium density as a uniform mixture of N Gaussian components with centers $\{(c_j^1, c_j^2)\}_{j=1}^N$ are sampled along the curve $(c_j^1, c_j^2) = (\sin t_j, \sin 2t_j)$ for $t_j = 2\pi j/N$. Because each component is separable, we can write this distribution as

$$\hat{\rho}(z) = \frac{1}{N}\sum_{i=1}^{N} \exp\left(-\frac{(z_1 - c_i^1)^2}{\sigma^2}\right) \exp\left(-\frac{(z_2 - c_i^2)^2}{\sigma^2}\right), \tag{24}$$

meaning that $\hat{\rho}$ has a rank-N TT representation of the form eq. (18) (critically, this differs from the double-well example because it is not exactly separable). The two metastable states are then taken to be Gaussians centered at $A = (-1, 0)$ and $B = (1, 0)$, which serve as the two end states of the figure-8 transition pathway. Equivalently, one may view these as the low-free-energy regions selected to define the committor boundary conditions, with effective free energy $F(z) = -\beta^{-1} \log \hat{\rho}(z)$.

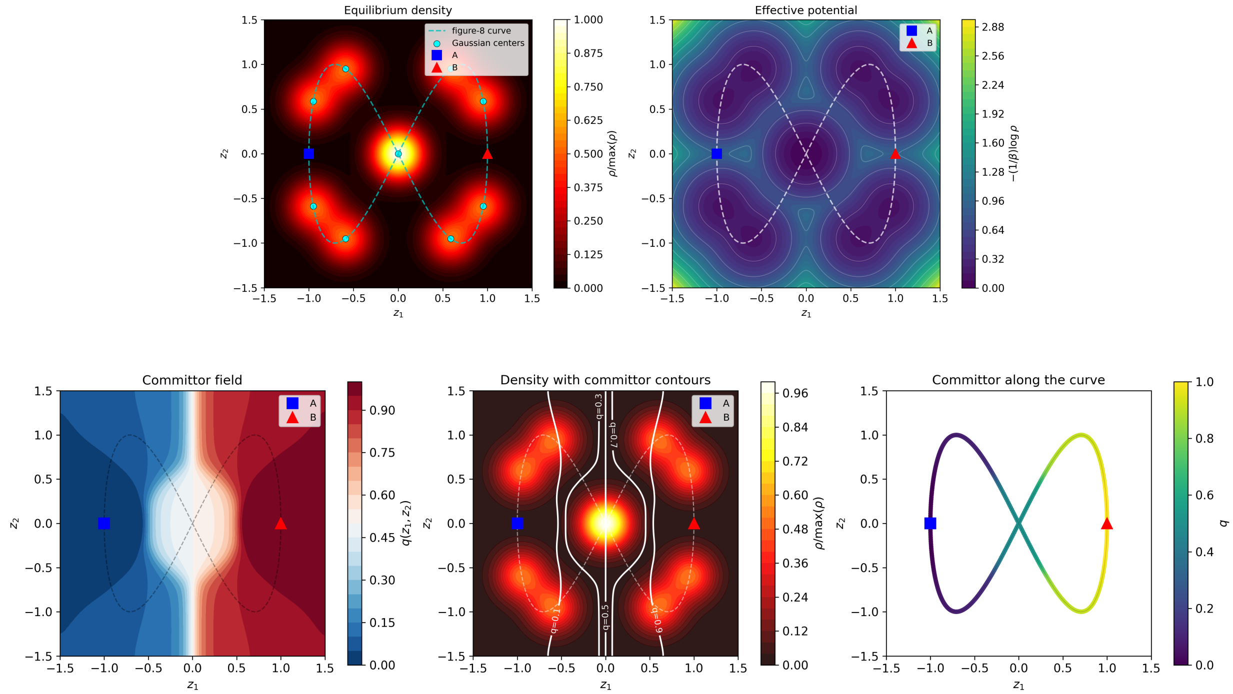

FIGURE 2. The 2 dimensional toy example with $N = 10$. In the top panel we simply plot eq. (24). In the bottom panel we plot the committor function values obtained using the TT algorithm. Qualitatively, these plots agree with what one would generally expect.

While the two-dimensional version of this toy problem is primarily useful as a sanity check, the more interesting case is the higher-dimensional extension. In two dimensions, this problem can still be solved by classical numerical approaches, so the TT-based formulation is not yet being pushed to its limits. The four-dimensional setting is already much more representative of the high-dimensional regimes that motivate RECOVAR-like latent spaces, and it is here that tensor-train methods become genuinely useful. In this case we consider the density

$$\hat{\rho}(z) = \frac{1}{N}\sum_{i=1}^{N} \exp\left(-\frac{(z_1 - c_i^1)^2}{\sigma^2}\right) \exp\left(-\frac{(z_2 - c_i^2)^2}{\sigma^2}\right) \exp\left(-\frac{(z_3 - c_i^3)^2}{\sigma^2}\right) \exp\left(-\frac{(z_4 - c_i^4)^2}{\sigma^2}\right). \tag{25}$$

We were able to implement this extension and obtain qualitatively reasonable results, which suggests that the example does capture some of the intended geometry of a low-dimensional conformational manifold embedded in a higher-dimensional state space.

That said, this implementation should be interpreted cautiously. In its present form, it does not scale as cleanly as one would hope. The computation was expensive and, in several runs, took significantly longer than expected to converge. This appears to reflect the fact that the example is somewhat delicate to construct directly in tensor-train form, especially without a dedicated TT approximation procedure. Nevertheless, the resulting density and committor profiles look broadly consistent with the intended figure-8 transition picture, so we include them here as a proof-of-concept rather than as a fully optimized benchmark.

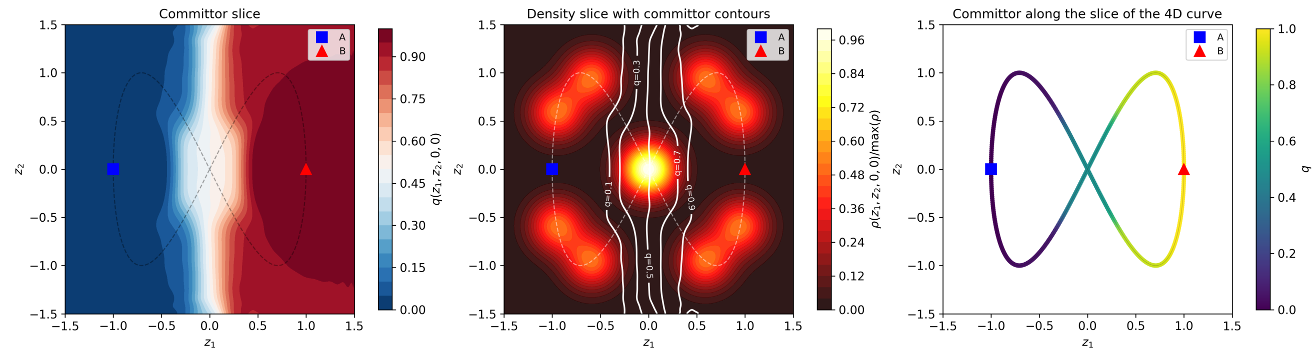

FIGURE 3. The 4 dimensional toy example with $N = 10$. Qualitatively, these plots look pretty similar to the two dimensional case and so agree with what one would generally expect. We note that this result was highly sensitive to parameter selection so we hypothesize that better results can be obtained with more careful parameter tuning.

5. FUTURE

As discussed in Section 2.2, RECOVAR must perform a density estimation in latent space to obtain the conformational density. Though we referred to this latent space as "low-dimensional," concretely it tends to be somewhere around dimension 4-10. Even in dimension 4, however, parametric density estimation algorithms, such as kernel density estimation (KDE), begin to fail because of what is referred to as *the curse of dimensionality*. This very general problem of using non-parametric methods to perform high-dimensional density estimation has been studied extensively. One method that we note for its connection to how we obtain the committor is [4] (see, e.g. `https://github.com/matias-ma/tt-de` for an implementation of this). This algorithm essentially assumes that the underlying density has a TT expansion as in eq. (18) and then minimizes the negative log-likelihood by running a gradient descent on the coefficient cores. Not only does using such an algorithm result in a more accurate estimation of the conformational density, but it also means that p is *a priori* in TT format (i.e. the conformational density is output in the form eq. (18)).

First and foremost, this means that we would no longer need to find the TT approximation of the estimated p, avoiding additional projection error and numerical instability. Second, it allows all quantities required for the committor computation to be expressed as tensor contractions, rather than numerical approximations of high-dimensional integrals.

From this perspective, TT density estimation and TT committor computation form a compatible pipeline since both rely on the same low-rank tensor structure and can share basis functions, tensor cores, and contraction routines. This suggests that an approach to cryo-EM heterogeneity, where both the density and the committor are learned in this TT format, may provide both computational and statistical advantages over other approaches.

This reformulation also hints at the convergence properties of this algorithm used to obtain the committor function. The ALS scheme used to solve eq. (14) is guaranteed to converge to a stationary point within the chosen class of TT cores, but this guarantee is only meaningful if the target functions admit accurate low-rank approximations. In particular, if both the equilibrium density p and the committor q are well-approximated in TT format, then the overall procedure is consistent up to the chosen ranks. Thus, the key question is not separability in a strict sense, but rather the existence of *low-rank approximations* for the functions of interest.

While exact separability is a restrictive assumption, there is evidence that sufficiently regular functions admit efficient low-rank representations. In the bivariate case, results based on continuous LU decompositions (see, e.g., [5]) show that analytic functions can be approximated with errors that decay rapidly with the rank. This suggests that regularity conditions such as analyticity of the effective free-energy landscape may, in practice, justify a TT approximation. A proof of this claim under suitable assumptions is omitted here, and we defer a more detailed treatment to future work.

I would like to thank Marc Aurèle Gilles for giving me the idea for this application.

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